

UNIRATIONALITY OF EVERY SMOOTH $(2, 3)$ HYPERSURFACE IN $\mathbf{P}^2 \times \mathbf{P}^2$ BY TANGENT RESIDUALS AND LATTÈS VARIATION: THE CONSTRUCTION AND ITS VERIFICATION ARTIFACTS

ABSTRACT. Let $X \subset \mathbf{P}_x^2 \times \mathbf{P}_y^2$ be any smooth hypersurface of bidegree $(2, 3)$ over $\mathbf{C}$. We prove that X is unirational. A suitable vertical surface $S = X \cap (\mathbf{P}^2 \times L)$ is rational by Tsen's theorem and maps birationally, by fiberwise tangent residuals in the plane-cubic fibration, onto a horizontal divisor $T \sim aH_x + H_y$ with $1 \leq a \leq 10$. This gives a dominant rational parametrization of even degree $2a \leq 20$. The key uniform input is the quartic residual-line Lattès map of a smooth plane cubic: its critical-value curve is the dual sextic, so it recovers the embedded cubic and cannot remain constant in a smooth nonconstant family. For a general equation a universal resultant identity gives $T \sim 10H_x + H_y$ and degree exactly 20. Combined with known results, the construction yields smooth threefolds that are unirational and irrational, with very general members additionally not stably rational. These notes are self-contained on the mathematics and additionally document every machine computation the proof relies on, with pointers to the certificate files in the accompanying repository.

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1. INTRODUCTION AND STATEMENT

A *standard conic bundle* over $\mathbf{P}^2$ is a smooth projective threefold X with a flat morphism $\pi: X \rightarrow \mathbf{P}^2$ whose fibers are plane conics, with relative Picard rank one. Its discriminant $\Delta \subset \mathbf{P}^2$ is a curve; the birational geometry of X is largely governed by $\deg \Delta$. Rationality is classical: X is rational iff $\deg \Delta \leq 4$, or $\deg \Delta = 5$ with even theta-characteristic (Beauville, Shokurov; see [13] for a survey), while a conic bundle with $\deg \Delta \geq 6$ is irrational by the Chow/Prym criterion [2, Théorème 4.9]. *Unirationality* is far more delicate. The published record reaches discriminant degree 8: conic bundles over $\mathbf{P}^2$ with $\deg \Delta \leq 8$ are unirational (Kollár–Mella [9] together with [13, §14.3]; also the preprint [11], which further covers degree 9 with *singular* discriminant). For large general discriminants the expectation in the literature was negative: “*It is likely that a conic bundle with sufficiently large and general discriminant curve is not unirational*” [13, §14.3], echoed in [10].

Every smooth hypersurface X of bidegree $(2, 3)$ in $\mathbf{P}^2 \times \mathbf{P}^2$ is a standard conic bundle over the second factor with discriminant of degree 9; the general discriminant is smooth. This is precisely the first smooth-discriminant case beyond the earlier published boundary. These notes give a complete proof of:

Theorem 1.1. *Every smooth hypersurface $X \subset \mathbf{P}_x^2 \times \mathbf{P}_y^2$ of bidegree $(2, 3)$ over $\mathbf{C}$ is unirational. More precisely, there is a dominant rational map $\mathbf{P}^3 \dashrightarrow X$ of even degree at most 20.*

Theorem 1.2 (Generic degree). *For a general $(2, 3)$ equation, Construction 3.1 gives a dominant rational map $\mathbf{P}^3 \dashrightarrow X$ of degree exactly 20.*

Corollary 1.3. *Every smooth $(2, 3)$ hypersurface is unirational (Theorem 1.1) and irrational [2, Théorème 4.9]. A very general member is in addition not stably rational [3, Theorem 1.1]; for such a member, $\mathrm{Br}(X) = 0$ and $H_{\mathrm{nr}}^3(X, \mathbf{Q}/\mathbf{Z}(2)) = 0$ [5], so the failure of stable rationality is invisible to the classical unramified invariants; unlike the Artin–Mumford threefold [1], whose non-stable-rationality is detected by Br , these examples are detected only by degeneration of the diagonal [3].*

The construction is elementary in style and could have been carried out classically; the nearest antecedent is the 1980 note of Del Centina–Verdi [6], which asserted a *special* smooth unirational $(3, 2)$ example by an entirely different method (rationality of one double plane with special branch sextic). Their displayed data contain inconsistencies; a corrected pencil is certified in the repository (§5). The fiberwise tangent-residual mechanism used here appears to be new.

Provenance of these notes. The construction and proof below were developed in a computational project archived at <https://github.com/mattrobball/unirational>. Its finite algebraic identities were mechanically verified. The mathematical exposition is self-contained; §5 is a complete ledger of the machine artifacts the proof relies on, what each verifies, and which steps rest on standard prose mathematics instead.

2. SETUP

Convention 2.1. We work over $\mathbf{C}$ (any algebraically closed field of characteristic 0 works verbatim). Coordinates $x = (x_0:x_1:x_2)$ on $\mathbf{P}_x^2$ and $y = (U:V:W)$ on $\mathbf{P}_y^2$. A bidegree- $(2, 3)$ form is written

$$F(x, y) = x^\top A(y) x,$$

with $A(y)$ a symmetric 3×3 matrix of cubic forms in y . We write $X = \{F = 0\}$ and let H_x, H_y denote the restrictions of the two hyperplane classes. Unless a generic refinement is explicitly stated, X is assumed smooth. “General” means: in a dense Zariski-open subset of the parameter space $\mathbf{P}^{59} = \mathbf{P}H^0(\mathcal{O}_{\mathbf{P}^2 \times \mathbf{P}^2}(2, 3))$, intersected with the smooth locus.

The threefold X carries *two* fibrations:

- $\pi = \text{pr}_y: X \rightarrow \mathbf{P}_y^2$, a flat conic bundle with discriminant $\Delta = \{\det A(y) = 0\}$ of degree 9 (smooth for general F);
- $\rho = \text{pr}_x: X \rightarrow \mathbf{P}_x^2$, a fibration whose fiber over x is the plane cubic $C_x = \{F(x, -) = 0\} \subset \mathbf{P}_y^2$; its generic fiber is smooth by generic smoothness.

The interplay of the two is the whole story.

Lemma 2.2. *For smooth X : $\text{Pic}(X) = \mathbf{Z}H_x \oplus \mathbf{Z}H_y$ and $K_X = \mathcal{O}_X(-1, 0)$. Every effective divisor class $aH_x + bH_y$ has $a, b \geq 0$. Writing f_y (a conic fiber of π) and c_x (a cubic fiber of ρ) for the two curve classes,*

$$H_x \cdot f_y = 2, \quad H_y \cdot f_y = 0, \quad H_x \cdot c_x = 0, \quad H_y \cdot c_x = 3.$$

Consequently a surface $S \in |aH_x + bH_y|$ has degree $2a$ over $\mathbf{P}_y^2$ and degree $3b$ over $\mathbf{P}_x^2$. In particular every multisection of the conic bundle has even degree, and ρ has no rational section.

Proof. Grothendieck–Lefschetz for the ample divisor X in $\mathbf{P}^2 \times \mathbf{P}^2$ ($\dim X = 3$) gives Pic ; adjunction gives K_X . Effectivity: from $0 \rightarrow \mathcal{O}(a-2, b-3) \rightarrow \mathcal{O}(a, b) \rightarrow \mathcal{O}_X(a, b) \rightarrow 0$ and $H^1(\mathbf{P}^2 \times \mathbf{P}^2, \mathcal{O}(m, n)) = 0$ for all m, n , one gets $h^0(\mathcal{O}_X(a, b)) = h^0(\mathcal{O}(a, b)) - h^0(\mathcal{O}(a-2, b-3))$, nonzero only if $a, b \geq 0$. The intersection numbers are immediate. The parity statement is consistent with, and geometrically explains, the nontriviality of the 2-torsion Brauer class of the generic conic (restriction–corestriction). $\square$

Remark 2.3. Since every surface in the threefold X is a divisor, Lemma 2.2 constrains all of them. Adjunction gives $K_S = ((a-1)H_x + bH_y)|_S$ for $S \in |aH_x + bH_y|$; for a *horizontal* class ($a \geq 1$) this is effective or trivial, so *no smooth member of any horizontal linear system is rational*. Any horizontal surface with rational normalization — which Theorem 1.1 requires, by Proposition 2.4 below — is necessarily a singular member of its system. This is the precise sense in which the problem resists naive searches, and why the construction below must produce a highly singular divisor.

Proposition 2.4 (Multisection principle). *Let $T \subset X$ be an irreducible surface dominating $\mathbf{P}_y^2$ with rational normalization $\tilde{T}$, of degree N over $\mathbf{P}_y^2$. Then X is unirational of degree N . Conversely, if X is unirational it contains such surfaces.*

Proof. Base-change: $X \times_{\mathbf{P}_y^2} \tilde{T} \rightarrow \tilde{T}$ is a conic bundle over a rational surface carrying the tautological section $t \mapsto (t, t)$. A conic with a rational point over the field $\mathbf{C}(\tilde{T})$ is isomorphic to $\mathbf{P}^1$, so $X \times_{\mathbf{P}_y^2} \tilde{T}$ is birational to $\tilde{T} \times \mathbf{P}^1$, a rational threefold; the projection to X is dominant of degree N . For the converse, the image of a general plane under a dominant map $\mathbf{P}^3 \dashrightarrow X$ is a horizontal surface with rational (indeed unirational, hence by Castelnuovo rational) normalization. $\square$

Harris’s formulation of the problem — find rational surfaces in X *other than* the preimages of rational curves in $\mathbf{P}_y^2$ [8, p. 24] — asks exactly for the horizontality in Proposition 2.4: the preimage $\pi^{-1}(\text{curve})$ is vertical and useless. The point of the construction below is that one

of these “useless” vertical surfaces can be *converted* into a horizontal one using the second fibration ρ .

3. THE GENERIC DEGREE-20 CONSTRUCTION

In this section F is general. The universal identity survives every specialization; Section 4 will use its full line-space variation to remove this generality assumption.

Construction 3.1. Fix a general line $L \subset \mathbf{P}_y^2$ and choose coordinates with $L = \{W = 0\}$.

- (I) **The vertical seed.** Let $S = X \cap (\mathbf{P}_x^2 \times L) \in |\mathcal{O}_X(0, 1)|$. Projection to $L \simeq \mathbf{P}^1$ makes S a conic bundle over $\mathbf{P}^1$; its generic fiber is a smooth conic over the C_1 field $\mathbf{C}(L) \simeq \mathbf{C}(t)$, hence has a rational point by Tsen’s theorem, and S is a *rational* surface (irreducible, since the generic fiber is geometrically integral). Under the other projection, S is a *trisection* of ρ : for general x , the fiber is $C_x \cap L = \{p_1, p_2, p_3\}$.
- (II) **Tangent residuals.** For general x the cubic C_x is smooth at each p_i (this needs only that the binary cubic $g_x = F(x, -)|_L$ has distinct roots). Define

$$\tau: S \dashrightarrow X, \quad (x, p) \mapsto (x, \tau_x(p)), \quad T_p C_x \cdot C_x = 2p + \tau_x(p).$$

In the group law of C_x (any flex as origin) this is $p \mapsto -2p$.

- (III) **The horizontal image.** Let $T = \overline{\tau(S)}$. Theorem 3.4 below shows $T \in |10H_x + H_y|$ and $\tau: S \dashrightarrow T$ is birational. By Lemma 2.2, T has degree $T \cdot f_y = 20$ over $\mathbf{P}_y^2$; it is horizontal with rational normalization, so Proposition 2.4 gives a dominant degree-20 map $\tilde{T} \times \mathbf{P}^1 \dashrightarrow X$, proving Theorem 1.2.

Everything therefore rests on identifying the image divisor exactly. This is done by a universal polynomial identity.

3.1. The universal resultant identity. Write a ternary cubic in the coordinates adapted to $L = \{W = 0\}$ as

$$(1) \quad f(U, V, W) = g(U, V) + W h(U, V) + W^2(iU + jV) + kW^3,$$

with g a binary cubic and h a binary quadratic. For $p = [u:v:0]$ with $g(u, v) = 0$, the tangent line to $\{f = 0\}$ at p , evaluated at $[U:V:W]$, is

$$P_{U,V,W}(u, v) = U g_u(u, v) + V g_v(u, v) + W h(u, v),$$

because f_U, f_V, f_W restrict on $W = 0$ to g_u, g_v, h . Set

$$\mathcal{R}_f(U, V, W) = \text{Res}_{u,v}(g, P_{U,V,W}),$$

the resultant of binary forms of degrees 3 and 2; up to the standard normalization, $\mathcal{R}_f$ is the product of the three tangent lines at the points of $\{g = 0\}$, a cubic in (U, V, W) .

Theorem 3.2 (Universal identity). *Let $\Delta(g)$ denote the classical discriminant of the binary cubic $(18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2)$ for $g = au^3 + bu^2v + cuv^2 + dv^3$. Then, identically in the ten coefficients of f ,*

$$(2) \quad \mathcal{R}_f + \Delta(g) f = W^2 q_f,$$

where q_f is a linear form in (U, V, W) whose three coefficients are homogeneous of degree 5 in the coefficients of f and have greatest common divisor 1 in the universal coefficient ring.

Proof. Expand the left side in W . The W^0 term is $\text{Res}(g, Ug_u + Vg_v) + \Delta(g)g(U, V)$. By Euler's relation $ug_u + vg_v = 3g$, each factor $Ug_u(p_i) + Vg_v(p_i)$ of the resultant is a linear form vanishing at the root p_i , so the product is proportional to $g(U, V)$; the constant is $-\Delta(g)$ (a classical covariant identity, verified symbolically in the artifacts). The W^1 coefficient of $\mathcal{R}_f$ is a binary quadratic in (U, V) ; evaluating at a root p_k of g kills every term of the root-expansion except the one attached to p_k , and comparison at the three roots identifies it with $-\Delta(g)h(U, V)$ (two binary quadratics agreeing at three distinct points are equal; the case $\Delta(g) = 0$ follows by continuity of a polynomial identity). Hence the left side of (2) is divisible by W^2 , and being cubic, the quotient q_f is linear. The coefficient degrees: $\text{Res}_{3,2}$ has bidegree (2, 3) in the coefficients of its two arguments, and the coefficients of P are linear in those of f , giving degree $2 + 3 = 5$; likewise $\deg \Delta(g) \cdot f = 4 + 1$. Primitivity ($\gcd = 1$) is a machine fact (§5, artifact A1/A4). $\square$

Remark 3.3 (Geometric content). Restricted to the cubic $C = \{f = 0\}$, the product of the three tangent lines has divisor $\sum_i (2p_i + r_i)$ where $r_i = \tau(p_i)$, while W has divisor $p_1 + p_2 + p_3$. On the cubic, $2p_i + r_i \sim H_y|_C$ and $\sum p_i \sim H_y|_C$, whence

$$r_1 + r_2 + r_3 \sim 3H_y|_C - 2H_y|_C = H_y|_C.$$

A degree-3 effective divisor on a smooth plane cubic lying in $|H_y|_C|$ is a line section: *the three tangent residuals are collinear*, and $\{q_f = 0\}$ is the line through them. (Group-law formulation: the residual map is $p \mapsto -2p$ and $(-2p_1) + (-2p_2) + (-2p_3) = -2(p_1 + p_2 + p_3) = 0$.)

3.2. The image divisor and its class. Now let the coefficients of $f = F(x, -)$ vary: they are *quadratic* forms in x . Then $\Delta(g_x)$ has bidegree (8, 0), $\mathcal{R}$ has bidegree (10, 3), and the covariant line $q = q_{F(x, -)}(y)$ has bidegree (10, 1) — linear in y , with three coefficient forms $q_U, q_V, q_W \in \mathbf{C}[x]_{10}$.

Theorem 3.4. *For general F :*

- (1) *On X the identity (2) reads $\mathcal{R} = W^2q$, giving the exact divisor equality*
 $\text{div}_X(\mathcal{R}) = 2S + T_q, \quad T_q := \text{div}_X(q) \in |10H_x + H_y|, \quad (10, 3) = 2(0, 1) + (10, 1).$
- (2) *T_q has no component vertical over $\mathbf{P}_x^2$; its fiber over a general x is exactly the three distinct tangent residuals $\{r_1, r_2, r_3\}$, reduced.*
- (3) *$T_q = T = \overline{\tau(S)}$ is irreducible and $\tau: S \dashrightarrow T$ is birational.*

Proof. (1) $q|_X \not\equiv 0$ since bidegree (10, 1) admits no factor of bidegree (2, 3), and then $\mathcal{R}|_X = W^2q|_X \not\equiv 0$; W cuts S (reduced and irreducible for general F, L).

(2) A component of T_q vertical over $\mathbf{P}_x^2$ (i.e. of class $(a, 0)$) would force $q(x, -) \equiv 0$ along a curve of x 's, i.e. a common factor of q_U, q_V, q_W . For each potential common-factor degree $e \geq 1$, the locus of coefficient triples with such a factor is the image of the proper multiplication map $\mathbf{P}(\mathbf{C}[x]_e) \times \mathbf{P}(\mathbf{C}[x]_{10-e})^3 \rightarrow \mathbf{P}(\mathbf{C}[x]_{10})^3$ -incidence, hence closed; there are finitely many e ; so $\{F : \gcd(q_U, q_V, q_W) = 1\}$ is open. It is nonempty by an explicit witness (artifact A1/A4), hence dense. Then $q(x, -) \equiv 0$ for only finitely many x , and no vertical divisorial component exists. For the fiber: over general x , $C_x \cap \{q(x, -) = 0\}$ is three points by Bézout, and by Remark 3.3 the three residuals r_i lie there; that they are pairwise distinct and off L for general x is an open condition on F satisfied at an explicit witness (see Lemma 3.5 and artifact A1/A4), so the fiber is exactly $\{r_1, r_2, r_3\}$, reduced.

(3) $\overline{\tau(S)} \subseteq T_q$ by (2); both have degree 3 over $\mathbf{P}_x^2$ ($T_q \cdot c_x = 3$ by Lemma 2.2), and by (2) the generic fiber of T_q is exactly the injective image of the generic fiber of S . Hence T_q is the

closure of the image of the irreducible surface S (irreducible), and τ is generically injective of degree 1. $\square$

Lemma 3.5 (Openness of the product projection). *Let $\mathcal{U} \subset \mathbf{P}^{59} \times \mathbf{P}_x^2$ be the open locus of pairs (F, x) where g_x has distinct roots, $q_{F(x, -)} \neq 0$, the three residuals are pairwise distinct, and none lies on L . Then $\{F : \exists x, (F, x) \in \mathcal{U}\}$ is open in $\mathbf{P}^{59}$, because the product projection $\mathbf{P}^{59} \times \mathbf{P}_x^2 \rightarrow \mathbf{P}^{59}$ is flat and locally of finite presentation, hence open. A single explicit pair $(F_0, x_0) \in \mathcal{U}$ therefore makes this open nonempty, hence dense. Concretely, all four conditions are checked at a witness by: $\text{disc}(g_{x_0}) \neq 0$; $q \neq 0$; $\text{disc}(F(x_0, -)|_{\{q=0\}}) \neq 0$ (three distinct residuals, no root extraction needed); and $F(x_0, -) \neq 0$ at the point $\{q=0\} \cap L$. $\square$*

Remark 3.6 (Independent Chow-theoretic check of the class). In $\mathbf{P}_x^2 \times L_p \times \mathbf{P}_r^2$ with hyperplane classes H, P, R , the incidence $\{p \in C_x \cap L, r \in T_p C_x \cap C_x\}$ is a complete intersection of classes $(2H+3P), (2H+2P+R), (2H+3R)$; pushing down L_p gives $20H^2 + 36HR + 9R^2$. Subtracting twice the pushed diagonal (tangency multiplicity 2) leaves

$$20H^2 + 32HR + 3R^2 = (2H + 3R)(10H + R),$$

i.e. the residual cycle is a divisor of class $(10, 1)$ on the $(2, 3)$ hypersurface — the same answer by an independent route.

3.3. The explicit parametrization. For computations (and for the external verification of §5) it is convenient to have the composite map in closed staged form. With $L = \{W = 0\}$, parametrize L by $y(t) = (t:1:0)$, and suppose (after the harmless linear specialization described in §5, A4) that $e = (1:0:0)$ lies on every conic $F(-, y(t))$. Writing $A_{ij}(t)$ for the restricted coefficients of the conic:

$$\begin{aligned} \text{(I)} \quad x^*(t, m) = & (A_{11} + mA_{12} + m^2A_{22}, \\ & -(A_{01} + mA_{02}), -m(A_{01} + mA_{02})) \end{aligned}$$

This is the second point where the line through e in direction $(0, 1, m)$ meets the conic. Hence it birationally parametrizes S by (t, m) .

$$\begin{aligned} \text{(II)} \quad f^* = & F(x^*(t, m), -), \quad (\alpha, \beta, \gamma) = \nabla_y f^*|_{y(t)}, \quad z^* = (\gamma, 0, -\alpha), \\ f^*(sy(t) + uz^*) = & u^2(As + Bu), \quad r(t, m) = By(t) - Az^* \end{aligned}$$

(z^* is a second point of the tangent line, since $\alpha\gamma + \gamma(-\alpha) = 0$; the two leading coefficients vanish identically, expressing tangency, and r is the residual).

$$\begin{aligned} \text{(III)} \quad d = & (0, 1, n), \quad Q = F(-, r), \\ B_Q(u, v) = & Q(u + v) - Q(u) - Q(v), \\ x'(t, m, n) = & Q(d)x^* - B_Q(x^*, d)d. \end{aligned}$$

Then $\Phi(t, m, n) = (x'(t, m, n); r(t, m))$ is a dominant rational map $\mathbf{A}^3 \dashrightarrow X$ of degree 20. Stage (III) is the chord construction on the conic fiber through the marked point x^* , and its correctness is the one-line *reflection identity* for any ternary quadratic form Q with polarization B_Q :

$$Q(Q(d)x - B_Q(x, d)d) = Q(d)^2 Q(x),$$

so $F(x', r) = Q(d)^2 F(x^*, r) = 0$. The degree over X equals $\deg(T/\mathbf{P}_y^2) = T \cdot f_y = 20$.

Remark 3.7 (Scope). The construction uses the plane-cubic fibration ρ , which exists precisely because the second degree is 3. It therefore says nothing about arbitrary standard conic bundles with smooth degree-9 discriminant (the matrix entries need not be all cubic), nor

about bidegrees $(2, d)$, $d \geq 4$, where the fibers of ρ have genus > 1 and the tangent line meets the fiber in $d - 2 > 1$ residual points (a correspondence, not a map). Harris's question for d large [8, p. 24] remains open.

Remark 3.8 (Byproducts in the repository). Two independent negative results certified in the same project delimit how sharp the construction is: on a general X , *every* member of $|H_x|$ is a $K3$ double plane (its branch sextic never acquires a point of multiplicity ≥ 3), and *no* member of $|H_x + H_y|$ has rational normalization. So the minimal-class searches fail provably, and the first horizontal rational surface found lives in $|10H_x + H_y|$ — a singular member, as forced by adjunction (Lemma 2.2 ff.).

4. REMOVING THE GENERALITY HYPOTHESIS

We now prove Theorem 1.1. The missing uniform point is not the resultant identity, which is universal, but horizontality: after a special specialization the three degree-10 coefficients of the residual line may have a common factor, and for a poorly chosen L the residual line could be constant in x . Varying L in the full dual plane rules this out.

4.1. The residual-line Lattès map. Let $C \subset \mathbf{P}^2$ be a smooth plane cubic. We work over a field of characteristic zero. Write

$$T_p C \cdot C = 2p + g(p).$$

After choosing a flex as origin, $g(p) = -2p$. Thus g is a finite morphism of degree 4 and preserves collinearity. If $L \cdot C = p + q + r$, the three points $g(p), g(q), g(r)$ are again collinear. They define a morphism

$$(4.1) \quad \delta_C : (\mathbf{P}^2)^\vee \longrightarrow (\mathbf{P}^2)^\vee, \quad L \longmapsto \overline{g(p)g(q)g(r)}.$$

Lemma 4.1 (Critical values). *The morphism δ_C has algebraic degree 4, and its reduced critical-value curve is the dual sextic $C^\vee$. In particular δ_C determines the embedded cubic C by biduality.*

Proof. Over an algebraic closure choose a flex as origin. The morphism

$$\pi : C \times C \longrightarrow (\mathbf{P}^2)^\vee, \quad (p, q) \longmapsto \overline{pq},$$

with the tangent line as its value on the diagonal, is the quotient by the natural S_3 -action on the triple $(p, q, -p - q)$. Its ramification divisor is the union of the three reflection curves on which two entries coincide, and its branch curve is $C^\vee$. Put $\mu = [-2] \times [-2]$. Then

$$(4.2) \quad \pi\mu = \delta_C\pi, \quad \pi^*R_{\delta_C} = \mu^*R_\pi - R_\pi = \sum_{i < j} \sum_{0 \neq \varepsilon \in C[2]} \{p_i - p_j = \varepsilon\}.$$

The curves on the right map to, and sweep out, the locus where two residual points coincide, namely $C^\vee$. This proves the critical-value assertion. Since $[-2]^*\mathcal{O}_C(1)$ has degree four times that of $\mathcal{O}_C(1)$, the induced plane map has algebraic degree 4. Finally $(C^\vee)^\vee = C$.

This construction and the critical-value statement also appear explicitly in [12, p. 964, (2.6)]; see also [7, §4.3]. The quotient proof above is algebraic and therefore applies over any characteristic-zero extension field. $\square$

Lemma 4.2 (Constant-value descent). *Let $K = \mathbf{C}(s, t)$ and $\varphi : \mathbf{P}_K^n \rightarrow \mathbf{P}_K^m$ be a morphism. If $\varphi(z) \in \mathbf{P}^m(\mathbf{C})$ for a Zariski-dense set of $z \in \mathbf{P}^n(\mathbf{C})$, then φ is defined over $\mathbf{C}$.*

Proof. Pass to source and target affine charts retaining a dense subset. Write the coordinate ratios as $R_i(z) \in K(z_1, \dots, z_n)$. Applying either $\mathbf{C}$ -derivation of K shows that $D(R_i)$ vanishes at a dense set of constant points, hence vanishes identically. The common constants of these derivations in $K(z_1, \dots, z_n)$ are $\mathbf{C}(z_1, \dots, z_n)$, so all ratios descend. The descended coordinate forms have no common zero because their base change is the given morphism. $\square$

4.2. Choosing a horizontal line section. Let X be an arbitrary smooth $(2, 3)$ hypersurface and put

$$K = \mathbf{C}(\mathbf{P}_x^2), \quad C = X_K \subset \mathbf{P}_{y,K}^2.$$

The generic cubic C is smooth by generic smoothness. For completeness, smoothness of X rules out a fiber equal to the whole $\mathbf{P}_y^2$: at such a fiber the two derivatives in independent local x -directions are plane cubics and have a common zero, which would be singular on X .

The embedded cubic C is not constant. Otherwise its equation over K would be $\lambda(x)f_0(y)$ for a fixed cubic f_0 . Comparing one nonzero coefficient makes λ a quadratic form, so $F(x, y) = Q(x)f_0(y)$ in $\mathbf{C}[x, y]$, contradicting smoothness.

Apply Lemma 4.1 to C/K . The morphism δ_C cannot be defined over $\mathbf{C}$: if it were, its critical-value sextic and then its bidual C would be constant. By Lemma 4.2, the constant lines having constant image under δ_C are not Zariski dense. We may therefore choose a constant line $L \subset \mathbf{P}_y^2$ such that

$$(4.3) \quad \delta_C(L) \notin (\mathbf{P}_y^2)^\vee(\mathbf{C}),$$

and simultaneously impose the following open conditions: L is not contained in the conic discriminant, $C \cap L$ is reduced, and $[-2]$ is injective on its three points. The last bad locus is, after algebraic closure, the finite union of curves swept out by the chords $\overline{p, p + \varepsilon}$ for $0 \neq \varepsilon \in C[2]$; the union is Galois invariant. A proper closed subset of the constant dual plane base-changed to K , together with these proper K -closed loci, cannot contain every constant point (expand its finitely many K -coefficients in a finite $\mathbf{C}$ -linearly independent set). Hence such an L exists.

The surface

$$S_L = X \cap (\mathbf{P}_x^2 \times L)$$

is integral: its generic fiber over L is a geometrically integral conic, and smoothness of X also rules out a whole $\mathbf{P}_x^2$ fiber by the same two-derivative argument. Tsen's theorem then gives $\mathbf{C}(S_L) = \mathbf{C}(t)(z)$, so S_L is rational. Tangent residuals define $\tau_L : S_L \dashrightarrow X$. The chosen injectivity condition shows that its image T_L is irreducible, $S_L \dashrightarrow T_L$ is birational, and $T_L \rightarrow \mathbf{P}_x^2$ has degree 3.

Suppose T_L were not horizontal over $\mathbf{P}_y^2$. Its image cannot be a point, since every conic fiber is one-dimensional. If its image were a curve, write $[T_L] = aH_x + bH_y$ using Lemma 2.2. A general conic fiber outside that curve is disjoint from T_L , so $2a = 0$; degree 3 over $\mathbf{P}_x^2$ gives $3b = 3$. Thus $[T_L] = H_y$. The restriction sequence gives

$$H^0(X, \mathcal{O}_X(0, 1)) = H^0(\mathbf{P}_y^2, \mathcal{O}(1)),$$

so T_L would be the inverse image of a constant line M . On the generic cubic this says $\delta_C(L) = M$, contradicting (4.3). Hence T_L is horizontal.

Finally put $L = \{W = 0\}$ and use Theorem 3.2. The residual line has three degree-10 coefficients in x . After dividing their gcd of degree e , its primitive equation cuts a divisor

$$D \sim (10 - e)H_x + H_y$$

containing T_L as its unique component dominating $\mathbf{P}_x^2$. We have $[T_L] = aH_x + H_y$. Every effective prime divisor $AH_x + BH_y$ has $A, B \geq 0$, by intersection with general conic and cubic fibers. Comparing coefficients in D therefore gives

$$1 \leq a \leq 10 - e \leq 10.$$

The normalization of T_L is rational and its degree over $\mathbf{P}_y^2$ is $2a \leq 20$. Base change of the conic bundle to this normalization has the tautological point and is birational to $\hat{T}_L \times \mathbf{P}^1$. Projection back to X proves Theorem 1.1.

Remark 4.3 (A broader rational case). If an integral (2, 3) member has geometrically integral but singular generic cubic over $\mathbf{C}(\mathbf{P}_x^2)$, its unique geometric singular point is rational over that field. Projection from it makes the generic cubic rational, so $\mathbf{C}(X) = \mathbf{C}(\mathbf{P}_x^2)(t)$ and X itself is rational. Thus the smooth generic cubic is the hard case. Generically geometrically reducible or nonreduced cubic fibrations require separate arguments.

5. VERIFICATION ARTIFACTS

The all-smooth proof uses one machine-checked algebraic fact, the universal identity (Theorem 3.2); its remaining inputs are the prose Lattès, descent, and Picard arguments of Section 4. The sharper generic degree-20 statement additionally uses primitivity of q and the explicit witnesses of Lemma 3.5. These finite facts are machine-checked. This section is a complete ledger. All paths are relative to the repository <https://github.com/mattrobball/unirational>; logs of every run are committed beside the scripts.

A0 — the all-smooth certificate. The complete standalone proof, including the quotient ramification formula, constant-value descent, simultaneous choice of L , and degree bound, is `certificates/all_smooth_tangent_residual_theorem.md`.

A1 — the certificate of record.

`problems/B-conic-bundle-multisections/certificates/tangent_residual_local_checks.py` (with `.log`):

- verifies (2) *fully symbolically* over the universal ring $\mathbf{Z}[a_3, \dots, k]$ in ten independent coefficient symbols (the W^0, W^1 coefficients of $\mathcal{R} + \Delta f$ vanish identically; the quotient by W^2 is linear);
- verifies each coefficient of q_f is homogeneous of degree 5 and that the three have gcd 1 universally, and again after a recorded substitution by ten quadratic forms (so the general (2, 3) equation has primitive q);
- supplies the smooth-fiber witness of Lemma 3.5: an explicit (F_0, x_0) with the cubic smooth (unit Gröbner bases in three charts), three residual points computed, verified to lie on the cubic, on the tangent lines, on $\{q = 0\}$, off L , pairwise distinct;
- records the divisor arithmetic $(10, 3) - 2(0, 1) = (10, 1)$, degree 20.

The full prose proof of the generic refinement is `certificates/tangent_residual_theorem.md`.

A2 — independent re-derivation (audit). During the audit of this session, the identity was re-derived from scratch in a fresh SymPy session sharing no code with A1: the W^0 divisibility, the cofactor being the *classical* discriminant (both sign conventions tested), degree-5 homogeneity, and universal gcd = 1 were reconfirmed independently.

A3 — independent end-to-end geometric check (audit). On an explicit cubic with rational points on L ($g = (u - v)(u - 2v)(u + v)$, generic h, i, j, k), the three tangent residuals were computed *from the raw geometry* (tangent lines; factoring the double root out of the restricted cubic) and verified to be distinct, off L , and to lie on $\{q_f = 0\}$ — confirming the geometric interpretation, not merely the algebra. A second check drew a random $(2, 3)$ form over $\mathbf{Q}$ and verified the witness conditions of Lemma 3.5 at $x_0 = (1, 2, 3)$, plus $\gcd(q_U, q_V, q_W) = 1$ by restriction to two lines.

A4 — external random-coefficient instance.

`problems/B-conic-bundle-multisections/external_check/`: a full end-to-end falsification test on coefficients drawn from `/dev/urandom` (recorded), with four coefficients zeroed so that the section $e = (1:0:0)$ exists over $\mathbf{Q}$ and the composite map has exact rational coefficients (a codimension-4 linear slice needed only because Tsen’s theorem is not effective over $\mathbf{Q}$; over $\mathbf{C}$ the theorem needs no such conditioning). The harness `build_explicit_map.py` constructs the staged map of §3.3 and verifies, exactly:

check	statement	method
C1	$F(x^*, y(t)) \equiv 0$	symbolic
C2	tangency (two leading coefficients vanish)	symbolic
C3	$f^*(r) \equiv 0$	full 100×53 grid [†]
C4	$q_{f^*}(r) \equiv 0$	full 62×37 grid [†]
C5	reflection identity	generic symbols
C6	$F(\Phi) = 0$ at sample rational points	exact evaluation
C7	Jacobian rank 3 (dominance)	exact symbolic differentiation
C8	fiber over random y_0 : squarefree of degree 20	resultant

[†]Deterministic proofs: a polynomial of bidegree at most (D_1, D_2) vanishing at a $(D_1+1) \times (D_2+1)$ grid of distinct rational nodes is zero; the bidegree bounds $(99, 52)$ and $(61, 36)$ are derived in the harness.

All eight pass (`run_log.txt`); the explicit map instance is `map_instance.txt`, and `README_CHECK.md` gives CAS-agnostic instructions for third-party verification with no trust in the pipeline.

A5 — unconditioned instance (no slice). `external_check/unconditioned_check.py` (with `.log`) addresses the natural objection that the codimension-4 slice in A4 might matter: it runs the full mechanism on a fresh entropy draw with *no* coefficient zeroed. With no rational point available, it works in the cubic étale algebra $K = \mathbf{Q}[\theta]/(g_0)$, $g_0 = F(x_0, -)|_L$, where the three marked points live; reduction modulo the full cubic verifies each identity at all three conjugate points simultaneously. It checks: primitivity of q_F ; the witness conditions of Lemma 3.5; tangency, $F(x_0, r) = 0$, $q(r) = 0$, and $r \notin L$ (via $\text{Res}(g_0, r_W) \neq 0$) in K ; the stage-(III) point $F(x', r) = 0$ in K ; and the squarefree degree-20 fiber. All pass. The slice in A4 is thereby confirmed to play no role beyond supplying $\mathbf{Q}$ -rationality of the composite parametrization (Tsen’s theorem being non-effective over $\mathbf{Q}$); by $\text{PGL}_3 \times \text{PGL}_3$ -equivariance the sliced family $\{X \supseteq \{e\} \times L\}$ is in no special position for anything the proof uses.

Prose-only steps. For completeness: the steps of the proof resting on standard mathematics rather than computation are Tsen’s theorem, generic smoothness, Grothendieck–Lefschetz (Lemma 2.2), the group-law and Lattès quotient arguments, constant-value descent, openness of the flat product projection in Lemma 3.5, and the Chow-theoretic cross-check. Each is supplied in the proof or cited to a primary source.

Independence and audit trail. A1 was rerun in a clean environment during the audit (Macaulay2 1.26.06 / SymPy 1.14.0) with output matching the committed logs; A2–A3 share no code with A1; A4 takes its input from an external entropy source. The literature verification (the boundary $\deg \Delta \leq 8$, the status of [11] as an unpublished preprint whose degree ≤ 8 consequence has published backing via [9, 13], the method and status of [6], and the residual-line Lattès statements in [12, 7]) was performed against primary sources on 2026-07-22.

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